A history of the telescope

17a

3 500 BC When Phoenicians were cooking on sand around 3 500 BC, they discovered glass. But it took thousands of years before glass was shaped into a lens for the first telescope. Lenses that could change the magnification of objects were known in Europe at the end of the thirteenth century.

1608 The spectacle maker Hans Lippershey (1570–1619), born in Germany and working in Holland, is often credited with the invention of the first telescope in 1608. If one placed a biconvex lens (a lens used to correct long-sightedness) about 40 cm away from a biconcave lens (a lens used to correct short-sightedness) and then looked through them, distant objects appeared closer. This spyglass was used for spotting the army's enemy.

1609 The telescope was introduced to astronomy in 1609 by the Italian scientist Galileo Galilei (1564–1642), who became the first man to see the craters of the moon and went on to discover sunspots, the four large moons of Jupiter and the rings of Saturn. Galileo's telescope was a simple refractor telescope, using a plano-convex lens and a plano-concave eye lens in a tube. This arrangement magnified up to 30 times but provided only a narrow field of view: Galileo could see no more than a quarter of the moon's face. The German Johannes Kepler (1571–1630) invented the form of the refracting telescope that is the basis for modern refractors; it has a convex eye lens. However, Kepler's telescope produced large inverted images with some chromatic distortions (the lenses acted like a prism and split the starlight into its constituent colours resulting in coloured fuzzy edges of the images).

1704 In 1704, Sir Isaac Newton announced a new concept in telescope design: instead of lenses, a curved mirror was used to gather in light and reflect it. This reflecting mirror acted like a light-collecting bucket. Newton's reflector telescope could magnify objects much more than could ever be obtained with a lens and without chromatic distortion. The reflecting telescope invented by Isaac Newton used a large spherical mirror at the end of a tube to reflect the image towards the open end where a small mirror tilted at 45° then projected the image to a convex eyepiece lens sitting on the side of the tube.

1974 Telescopes grew from the 15 cm mirror used by Newton to the 6 m mirror of the Special Astrophysical Observatory in Russia, which opened in 1974.

1930 Bernhard Schmidt (1879–1935) was born in Estonia, studied in Germany and worked as an optical technician at the Bergedorf Observatory, Hamburg. In 1930 he invented the telescope that is named after him. It uses a spherical mirror, not a parabolical reflector, and a correcting lens at the telescope opening to compensate for spherical aberration; thus it is a combination reflector-refractor system. The Schmidt telescope is the most popular among amateur astronomers because of its compact design and large aperture, and because the optics are completely enclosed. The largest Schmidt telescope is located at Palomar Observatory (California). It can photograph an area of the sky more than 300 times as large as that seen by other reflectors.

1977 Unfortunately the atmospheric layers have hampered telescope observations from the Earth's surface. Scientists, therefore, have developed other more efficient means for gathering data about the heavens. Some of the largest reflective telescopes are located high up on Palomar Mountain near San Diego, California, and on Mount Pastukhov in Russia. Scientists have carried telescopes in planes: the Kuiper Airborne Observatory, a cargo plane fitted with a 200 kg infrared telescope, is operated at about 15 km altitude. The Kuiper discovered the rings of Uranus in 1977 and the atmosphere around Pluto in 1988.

1990 Since 1990 the Hubble Space Telescope, run jointly by NASA and the European Space Agency, is orbiting the Earth at an altitude of 600 km and capturing images from what scientists believe is the edge of our universe.

1937 Radio waves are also a part of the electromagnetic spectrum. Just as an optical telescope gathers visible light and magnifies it, a radio telescope is a large dish-shaped device that gathers and amplifies radio frequencies from space. Radio telescopes do not 'see' the way optical telescopes do; the radio signals that the telescopes gather are used to create diagrams showing where the strongest radio sources are located. Grote Reber (1911–2002), an American amateur ham-radio enthusiast, built the world's first radio telescope in his backyard in Chicago, Illinois, in 1937. Reber's device was similar in design to the backyard satellite dishes of today, though of a considerably larger scale. If the dish is bigger, more data can be collected, and the resolution and sensitivity of the telescope is greater.

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1947 In 1947 Bernard Lovell (1913–) directed the construction of a parabolic radio reflector 66 m in diameter at Jodrell Bank, England. It was also there that a 76 m dish went into operation in August 1957.

1962 The 91m–diameter dish at Green Bank, West Virginia, built by the National Radio Astronomy Observatory in 1962, was the largest steerable radio telescope in the world, until it collapsed from metal fatigue in 1989.

1963 The largest individual dish currently in existence is in Arecibo, Puerto Rico. This radio telescope was created out of a natural depression within a ring of mountains in 1963. With a diameter of 305 m, the telescope's reflecting surface is too large to be steerable. Instead, a movable feed suspended above the dish provides access to most of the sky.

Perform the tasks below and answer the following questions in your science book.

Sketches

1 Sketch the following lenses: plano-convex, plano-concave, biconvex and biconcave.

2 Sketch the following mirrors: plane, spherical and paraboloid.

3 Sketch the telescopes built by Lippershey, Galilei, Kepler, Newton and Schmidt. Pay particular attention to the shape of the lenses and mirrors, as much as you can.

4 Sketch Reber's radio telescope.

On the lines

5 When did lenses appear first?

6 When and where was the first telescope built?

7 About how long was Lippershey's telescope?

8 What was the first telescope used for?

9 What is a refractor telescope made of?

10 How many times did Galilei's best telescope magnify heavenly objects?

11 How did Newton's telescope present a new concept?

12 What is chromatic distortion?

13 Why was Newton's telescope superior to Kepler's?

14 What was the advantage of Schmidt's telescope over Newton's?

Between the lines

15 How long ago was glass discovered?

16 Where did Lippershey get his lenses from for the first telescope?

17 As Galilei's telescope magnified more and more, it had to become longer and longer. What restriction resulted from having better magnification?

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18 How is Kepler's telescope different to Galilei's?

19 How did Newton overcome the problem of chromatic distortion?

20 How many years passed between the invention of the refractor telescope and the invention of the reflector telescope?

21 How many times is the mirror of the Special Astrophysical Observatory in Russia bigger than the one in Newton's telescope?

22 How is Schmidt's telescope a reflector-refractor telescope?

23 Why are today's telescopes mounted at high altitudes?

24 If larger telescopes produce better images, why don't scientists build them in kilometre sizes?

Beyond the lines

25 Where did the Phoenicians live?

26 If Galilei was the first person to see the craters on the moon, what did people before him see on the moon?

27 Galilei found that Jupiter had moons orbiting it. Why would this have caused interest among scientists and people? (Hint: People then believed in a geocentric universe.)

28 Newton's telescope had a large curved mirror and a small tilted one. Why was this second mirror small and tilted?

29 What in the atmospheric layers might hamper astronomical observations?

30 Why were Galilei, Kepler and Newton not concerned about the interference of the atmosphere during their telescope observations?

31 The Hubble Space Telescope sent pictures to Earth that were slightly out of focus, because the mirror was too flat at one edge by 1/50 of the width of a human hair. Why didn't the scientist notice this fault before launching the telescope into space?

32 How would the Hubble Space Telescope be repaired, if there were problems with its operation?

33 What else is part of the electromagnetic spectrum apart from light and radio waves?

34 How are the different wave types of the electromagnetic spectrum different and how are they similar?

Developing a timeline

25 On a separate sheet of paper, draw a timeline that can show all the dates in the three pages of this worksheet. Label it using thousands of years. Then add to it the names and short terms representing what you think are the major events in the history of the telescope.

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